AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271 Machine Learning and Deep Learning

Midterm Examination (MTA) Meke Up
Spring Semester 2024/2025

Date: 24/April/2025

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed: 120 Minutes**

Instructions:

- Total Marks: 30
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

ACCEPT

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Part – A (5 Marks)

Question No 1) Select the correct choice for each of the following statements.

1. Which of the following is a type of Machine Learning?
   A) Supervised Learning
   B) Unsupervised Learning
   C) Reinforcement Learning
   D) All of the above
2. What is the primary objective of unsupervised learning?
   A) To predict outputs based on labeled data
   B) To discover hidden patterns and structures in data
   C) To minimize classification error
   D) To improve reinforcement learning algorithms
3. Which machine learning technique is best suited for classifying categorical data?
   A) Regression
   B) Clustering
   C) Classification
   D) Reinforcement learning
4. Which of the following is the disadvantage of Decision Trees?
   A) Prone to overfitting
   B) Requires large amounts of data
   C) Difficult to interpret results
   D) Cannot handle categorical variables
5. In a Support Vector Machine (SVM), what is the role of a "kernel function"?
   A) To compute the margin width
   B) To transform data into a higher-dimensional space
   C) To reduce the number of support vectors
   D) To increase the number of data points

Part – B (15 Marks) Provide brief answers to the following questions.

Question No 2) Explain how a reinforcement learning model interacts with its environment? (3 Marks)

Question No 3) Explain the difference between supervised and unsupervised learning with examples. (6 Marks)

Question No 4) Explain the working of Gradient descent. What are its advantages? (6 Marks)

Part – C (10 Marks)

NOTE: This question does not need any programming code.

A retail company wants to analyze customer purchasing behavior to improve its marketing strategies. The company has a large dataset containing customer demographics, past purchase history, browsing behavior, and product preferences. Their goal is to segment customers into different groups to offer personalized promotions and recommendations.

Instructions:

Question No 5) **Problem Definition & Technique Selection:** Clearly define the problem and choose the most suitable machine learning approach (supervised, unsupervised, or another method). Justify your choice.

Question No 6) **Implementation Steps:** Outline the key steps in implementing the chosen machine learning algorithm.

Question No 7) **Model Evaluation & Optimization:** Explain how you would measure the performance of your model and optimize it for better results.

End of questions

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